INTEGRATION QUIZ

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Name: Solutions

1. Use right endpoints and the formulas

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \qquad \sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

to evaluate the definite integral

$$\int_0^2 x^3 \, \mathrm{d}x$$

$$\Delta x = \frac{2-0}{n} = \frac{2}{n}, \quad x_i = 0 + i\left(\frac{2}{n}\right) = \frac{2i}{n}, \quad f(x) = x^3$$

$$f(x_i)\Delta x = \left(\frac{2i}{n}\right)^3\left(\frac{2}{n}\right) = \frac{2^4 i^3}{n^4}$$

$$R_n = \sum_{i=1}^{n} f(x_i)\Delta x = \sum_{i=1}^{n} \frac{2^4 i^3}{n^4} = \frac{2^4}{n^4}\sum_{i=1}^{n} i^3$$

$$= \frac{2^4}{n^4}\left(\frac{n(n+1)}{2}\right)^2 = \frac{2^4}{2^2}\,\frac{n^2(n+1)^2}{n^4} = 2^2\,\frac{n^2+2n+1}{n^2}$$

$$= 4\left(1 + \frac{2}{n} + \frac{1}{n^2}\right)$$

$$\int_0^2 x^3 dx = \lim_{n\to\infty} R_n = \lim_{n\to\infty} 4\left(1 + \frac{2}{n} + \frac{1}{n^2}\right) = 4(1+0+0) = \boxed{4}$$